

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE FOR COMPOSITION-CONTROLLED SYNTHESIS, LATEX PROCESSING, AND POST-MODIFICATION

1. ABSTRACT

This report synthesizes four primary patents relevant to low-acrylonitrile nitrile butadiene rubber (NBR) and closely related nitrile-group-containing elastomers. The target material is defined as an NBR copolymer with reduced acrylonitrile content, typically below about 25 wt% and often in the 15–24 wt% range, where lower ACN improves low-temperature flexibility while retaining useful oil/fuel resistance. The selected patents span three distinct technical directions: direct NBR latex processing and coagulation control, controlled emulsion polymerization of nitrile-containing block copolymers, and post-polymerization hydrogenation of diene-based NBR latexes.

EP2238176B1 focuses on nitrile rubber containing acrylonitrile and 1,3-butadiene units, with very low chlorine content and improved latex coagulation behavior using polyvinyl alcohol (PVA) as a co-precipitant. The examples emphasize reduced salt demand for coagulation, rapid vulcanization, and low metal corrosivity of vulcanizates. EP3063192B1 discloses nitrile-group-containing copolymer rubbers with 10–60 wt% nitrile units, 31–89 wt% diene units, and 1–9 wt% acrylate-type termonomers, with hydrogenation levels of 50% or more; it is more aligned with HNBR-type materials than low-ACN NBR, but it provides composition-control and emulsion-processing insights. US7592409B2 teaches controlled emulsion polymerization using in-situ generated emulsifiers and controlled radical methods to make block copolymers containing acrylonitrile/isoprene or acrylonitrile/butadiene segments, demonstrating microemulsion formation, rapid conversion, and seed-latex block growth. US9657113B2 discloses direct hydrogenation of NBR latex using Ru/Os-based catalysts, showing that 66/34 butadiene/acrylonitrile latex can be hydrogenated efficiently in water without organic solvent.

Across the landscape, the dominant process themes are emulsion or latex-based synthesis, careful monomer composition control, low-salt or in-situ emulsifier strategies, and process conditions that preserve latex stability while enabling downstream vulcanization or hydrogenation. For low-ACN NBR synthesis, the strongest evidence supports emulsion polymerization with controlled feed/composition management, followed by coagulation routes that minimize ionic contamination and preserve cure performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2238176B1
- Patent Title: Nitrile rubbers which optionally contain alkylthio terminal groups and which ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses nitrile rubber based on acrylonitrile and 1,3-butadiene, prepared as an NBR latex and isolated by coagulation. The examples emphasize latex coagulation using magnesium chloride with polyvinyl alcohol (PVA) as a co-precipitant to reduce salt demand, followed by drying and evaluation of storage stability, scorch, vulcanization, and corrosion behavior. - Chlorine content: 4 to 25 ppm — Claim 1 - Chlorine content: 5 to 23 ppm — Claim 2 - Mooney viscosity (ML (1+4 @100°C)): 10 to 150 Mooney units — Claim 5 - Mooney viscosity (ML (1+4 @100°C)): 20 to 100 Mooney units — Claim 5 - Glass transition temperature: -70°C to +10°C — Claim 6 - Glass transition temperature: -60°C to 0°C — Claim 6 - Storage stability test: 48 hours at 100°C — Example I - Mooneyscorch test temperature: 120°C — Example III - Vulcanization rate test temperature: 160°C — Example III - Latex charge: 200 g — Example Series 1 - Latex temperature: 20°C — Example Series 1 - PVA solution concentration: 4% by weight in deionized water — Example Series 1 - MgCl₂ amount: 0.87 wt.% — Example Series 1 - Comparative MgCl₂ amount: 1.31 wt.% — Example Series 1 - Hydraulic pressure: 120 bar — Example 4 - Vulcanization time: 30 min — Example 4 - Vulcanization temperature: 160°C — Example 4 - Plate thickness: 2 mm — Example 5 - Vulcanization time: 15 min — Example 5 - Relative humidity: 80% — Example 5 - Storage temperature: 70°C — Example 5 - Storage duration: 6 weeks — Example 5

Relevant Experimental Evidence - Example Series 1: PVA reduced the minimum MgCl₂ required for quantitative latex coagulation compared with the no-PVA control. - Example Series 2: simultaneous MgCl₂ and PVA addition at 20°C followed by heating to 90°C was used to determine coagulation requirements. - Example Series 4: MgCl₂/PVA-coagulated NBR showed rapid vulcanization and good vulcanizate properties at 160°C. - Example Series 5: PVA-assisted coagulation enabled NBR with low metal corrosivity in aluminum contact testing.

Technical Relevance This patent is directly relevant because it concerns NBR latex processing and isolation, which are critical for low-ACN NBR manufacture. It also links composition/impurity

control to downstream vulcanization and corrosion performance, both important for rubber-grade low-ACN NBR.

Patent 2

- Patent Number: EP3063192B1
- Patent Title: Copolymer rubber containing nitrile groups
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses nitrile-group-containing copolymer rubbers comprising acrylonitrile or related nitriles, conjugated diene units, and a small amount of acrylate-type termonomer, with a hydrogenation level of 50% or more. The examples describe characterization of latexes and vulcanizates, and the synthesis context is centered on controlled preparation of hydrogenated nitrile copolymer rubbers rather than conventional low-ACN NBR. - α,β -ethylenically unsaturated nitrile unit: 10% to 60% by weight — Claim 1 - α,β -ethylenically unsaturated nitrile unit: 20% to 50% by weight — Claim 1 - α,β -ethylenically unsaturated nitrile unit: 34% to 43% by weight — Claim 1 - Conjugated diene unit: 31% to 89.0% by weight — Claim 1 - Conjugated diene unit: 42% to 78.75% by weight — Claim 1 - Conjugated diene unit: 49% to 64.5% by weight — Claim 1 - Termonomer content: 1% to 9% by weight — Claim 1 - Termonomer content: 1.25% to 8% by weight — Claim 1 - Termonomer content: 1.5% to 8% by weight — Claim 1 - Hydrogenation level: 50% or more — Claim 1 - Hydrogenation level: 75% or more — Claim 1 - Hydrogenation level: 85% or more — Claim 1 - Mooney viscosity (ML 1+4@100°C): 67 — HNBR 1 - ACN content: 33.6% — HNBR 1 - Methoxyethyl acrylate content: 4.8% — HNBR 1 - Mooney viscosity (ML 1+4@100°C): 88 — HNBR 2 - ACN content: 34.6% — HNBR 2 - Polyethylene glycol acrylate content: 4.1% — HNBR 2 - Mooney viscosity (ML 1+4@100°C): 72 — HNBR 3 - ACN content: 33.6% — HNBR 3 - Methoxyethyl acrylate content: 7.7% — HNBR 3 - ACN content: 34% — HNBR 4 - Residual double bond content: less than 0.9% — HNBR 4 - Mooney viscosity (ML 1+4@100°C): 70 — HNBR 4 - Roll mill diameter: 80 mm — Example 1 to 5 - Roll mill length: 200 mm — Example 1 to 5 - Roll temperature: 40°C — Example 1 to 5 - Front roll speed: 16.5 rpm — Example 1 to 5 - Rear roll speed: 20 rpm — Example 1 to 5 - Friction ratio: 1:1.2 — Example 1 to 5 - Total mixing time: 5 to 8 minutes — Example 1 to 5* - MDR test temperature: 180°C — Table 5 - MDR oscillation angle: 0.5° — Test methods - MDR oscillation frequency: 1.7 Hz — Test methods - Vulcanization time: 30 minutes — Test methods - Vulcanization temperature: 180°C — Test methods - Tg measurement range: -100°C to 100°C — Test methods - DSC heating rate: 20 K/minute — Test methods - Oil swelling test: 150°C for 7 days — Test methods - Hot air ageing: 168 h at 150°C — Table 8 - IRM 903 ageing: 168 h at 150°C — Table 10

Relevant Experimental Evidence - HNBR 1: *a hydrogenated nitrile-butadiene-methoxyethyl acrylate terpolymer with 33.6% ACN and 4.8% MEA was prepared.* - HNBR 2: *a hydrogenated nitrile-butadiene-poly(ethylene glycol) methyl ether methacrylate terpolymer with 34.6% ACN and 4.1% PEG5 was prepared.* - HNBR 3: *a second MEA-containing hydrogenated terpolymer with 7.7% MEA and 33.6% ACN was prepared.* - HNBR 4: *commercial HNBR with 34% ACN and less than 0.9% residual double bonds was used as a benchmark.* - Examples 1 to 5*: compounding and aging tests showed the effect of composition on vulcanization, tensile properties, Tg, and oil resistance.

Technical Relevance This patent is only indirectly relevant to low-ACN NBR because it centers on hydrogenated nitrile copolymers, but it provides composition windows and processing data for nitrile/diene elastomers. It is useful as a reference for how nitrile content, diene content, and monomer addition affect elastomer properties and downstream performance.

Patent 3

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses controlled emulsion polymerization using in-situ generated emulsifiers and controlled free-radical polymerization agents to make block copolymers in water. Relevant embodiments include block polymers where the B block contains acrylonitrile or methacrylonitrile with isoprene, and the examples demonstrate microemulsion formation, rapid conversion, and seed-latex growth of triblock and diblock architectures. - Latent surfactant amount: approximately 0.2 grams — Example 1 - Styrene amount: 5 milliliters — Example 1 - Surfactant activator solution volume: 5 milliliters — Example 1 - Styrene charge: 1,000 grams — Example 2 - Oleic acid: 60.0 grams — Example 2 - Dibenzyltrithiocarbonate: 7.2 grams — Example 2 - RO water: 4,000 grams — Example 2 - Potassium persulfate: 40.0 grams — Example 2 - Tripotassium phosphate: 40.0 grams — Example 2 - Potassium hydroxide: 16.4 grams — Example 2 - Reaction temperature: 65°C — Example 2 - Complete conversion time: less than 1.5 hours — Example 2 - Styrene charge: 1,000 grams — Example 3 - Hexadecyl half ester of maleic acid: 60.0 grams — Example 3 - Dibenzyltrithiocarbonate: 7.2 grams — Example 3 - RO water: 4,000 grams — Example 3 - Potassium persulfate: 40.0 grams — Example 3 - Tripotassium phosphate: 40.0 grams — Example 3 - Potassium hydroxide: 16.4 grams — Example 3 - Reaction temperature: 65°C — Example 3 - Complete conversion time: less than 3 hours — Example 3 - 0.6 phr K/Oleate soap solution: added before isolation — Example 3 - RO water: 200 grams — Example 4 - RO water: 241 grams — Example 5 - Reaction temperature: 75°C — Example 6 - Complete conversion time:

about 2 hours — Example 6 - Reaction temperature: 75°C — Example 8 - Complete conversion time: about 7 hours — Example 8 - Reaction temperature: 75°C — Example 10 - Complete conversion time: about 3.5 hours — Example 10 - Reaction temperature: 75°C — Example 13 - Complete conversion time: about 4 hours — Example 13 - Reaction temperature: 75°C — Example 14 - Complete conversion time: about 3 hours — Example 14 - Reaction temperature: 75°C — Example 15 - Complete conversion time: about 2 hours — Example 15 - Styrene charge: 2,240 grams — Example 16 - Oleic acid: 134.4 grams — Example 16 - Dibenzyltrithiocarbonate: 16.0 grams — Example 16 - RO water: 3,808 grams — Example 16 - Potassium persulfate: 13.44 grams — Example 16 - Tripotassium phosphate: 89.6 grams — Example 16 - Potassium hydroxide: 36.96 grams — Example 16 - Reaction temperature: 65°C — Example 16 - Solids: 38.9% — Example 16 - Styrene charge: 50 grams — Example 17 - Oleic acid: 3.0 grams — Example 17 - Sodium nitrite: 0.69 grams — Example 17 - Acetic acid: 1.0 grams — Example 17 - Water: 2 grams — Example 17 - RO water: 160 grams — Example 17 - Potassium persulfate: 2.0 grams — Example 17 - Tripotassium phosphate: 2.0 grams — Example 17 - Potassium hydroxide: 2.35 grams — Example 17 - Reaction temperature: 75°C — Example 17 - Complete conversion time: about 2.5 hours — Example 17 - Styrene charge: 50 grams — Example 18 - Oleic acid: 3.0 grams — Example 18 - 1-chloro-1,1-diphenylethane: approximately 0.01 moles — Example 18 - RO water: 200 grams — Example 18 - Potassium persulfate: 2.0 grams — Example 18 - Tripotassium phosphate: 2.0 grams — Example 18 - Potassium hydroxide: 1.55 grams — Example 18 - Reaction temperature: 75°C — Example 18 - Complete conversion time: about 4 hours — Example 18 - Styrene charge: 1,300 grams — Example 19 - Oleic acid: 78.0 grams — Example 19 - Crude 1-chloro-1,1-diphenylethane: 39.0 grams — Example 19 - Deionized RO water: 2,470 grams — Example 19 - Potassium persulfate: 50.7 grams — Example 19 - Tripotassium phosphate: 50.7 grams — Example 19 - Potassium hydroxide: 29.5 grams — Example 19 - Reaction temperature: 75°C — Example 19 - Complete conversion time: about 5.5 hours — Example 19 - 1-hexene: 25 grams — Example 20 - n-butyl acrylate: 25 grams — Example 20 - Dipyrityl: 0.38 grams — Example 20 - Oleic acid: 4.0 grams — Example 20 - Ethyl 2-bromoisobutyrate: 0.47 grams — Example 20 - 85% KOH: 1.7 grams — Example 20 - Water: 139 milliliters — Example 20 - Copper sulfate pentahydrate solution: 60.9 grams of 1% solution — Example 20 - Hydrazine hydrate: 2 drops — Example 20 - Reaction temperature: about 67°C — Example 20 - Reaction time: 1.5 hours — Example 20 - Final solids: 10.9% — Example 20

Relevant Experimental Evidence - Example 1: multiple latent surfactant/activator pairs were shown to generate emulsions in styrene. - Example 2: RAFT emulsion polymerization gave polystyrene latex with 54,000 Mn and PDI 1.17 in less than 1.5 hours. - Example 3: in-situ generated polymerizable surfactant enabled similar RAFT latex formation with 48,000 Mn and PDI 1.17. - Example 5: S-NBR-S triblock copolymer latexes were prepared from PS seed latexes. - Example 20: microemulsion ATRP of n-butyl acrylate and 1-hexene demonstrated the broader controlled-emulsion platform.

Technical Relevance This patent is relevant because it provides a controlled emulsion polymerization framework that can be adapted to nitrile/diene monomer systems. Its in-situ emulsifier and seed-latex concepts are especially useful for composition-controlled low-ACN NBR or NBR-like block architectures.

Patent 4

- Patent Number: US9657113B2
- Patent Title: Hydrogenation of a diene-based polymer latex
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses direct hydrogenation of diene-based polymer latexes in water using solid ruthenium- or osmium-based catalysts, without organic solvent. The examples specifically hydrogenate a butadiene-acrylonitrile copolymer latex containing 66% butadiene and 34% acrylonitrile, showing that high hydrogenation degrees can be achieved while maintaining latex stability and avoiding gel formation. - NBR solid content: 19.5 wt % in water — Table 1 - Butadiene content: 66% — Table 1 - Acrylonitrile content: 34% — Table 1 - Mean particle diameter: about 72 nm — Table 1 - NBR latex volume: 25 ml — Example 1 - Water volume: 75 ml — Example 1 - HG2 catalyst: 0.00487 g — Example 1 - Nitrogen degassing: about 200 psi (1.38 MPa) — Example 1 - Nitrogen degassing time: 20 min — Example 1 - Reaction temperature: 100°C — Example 1 - Hydrogen pressure: 1000 psi (6.89 MPa) — Example 1 - Hydrogenation time: 4 hours — Example 1 - Hydrogenation degree: 98% — Example 1 - G2 catalyst: 0.0066 g — Comparative Example 1a - Hydrogenation time: 10 hours — Comparative Example 1a - Hydrogenation degree: 22% — Comparative Example 1a - HG2 catalyst: 0.00244 g — Example 2 - Hydrogenation time: 19 hours — Example 2 - Hydrogenation degree: 90% — Example 2 - Reaction temperature: 120°C — Example 3 - Hydrogenation time: 1.6 hours — Example 3 - Hydrogenation degree: 98% — Example 3 - Catalyst loading: 0.00975 g — Example 4 - Hydrogenation time: 0.5 hours — Example 4 - Hydrogenation degree: 98% — Example 4 - Catalyst loading: 0.00125 g — Example 5 - Hydrogenation time: 10 hours — Example 5 - Hydrogenation degree: 74% — Example 5 - Hydrogen pressure: 500 psi (3.45 MPa) — Example 6 - Hydrogenation time: 10 hours — Example 6 - Hydrogenation degree: 99% — Example 6 - Hydrogen pressure: 1500 psi (10.34 MPa) — Example 7 - Hydrogenation time: 9 hours — Example 7 - Hydrogenation degree: 99% — Example 7 - NBR latex volume: 100 ml — Example 8 - HG2 catalyst: 0.01253 g — Example 8 - Hydrogenation time: 3 hours — Example 8 - Hydrogenation degree: 99% — Example 8 - Molecular weight of HNBR: 249,000 — Example 1 - PDI of HNBR: 3.1 — Example 1

Relevant Experimental Evidence - Example 1: 66/34 butadiene/acrylonitrile NBR latex was hydrogenated to 98% in 4 hours at 100°C and 1000 psi H₂ with HG2 catalyst. - Comparative Example 1a: Grubbs II catalyst was much less effective, reaching only 22% hydrogenation after 10 hours. - Example 3: raising temperature to 120°C shortened the time to 98% hydrogenation to 1.6 hours. - Example 6: lowering hydrogen pressure to 500 psi still gave 99% hydrogenation after 10 hours. - Example 8: undiluted latex also hydrogenated efficiently to 99% conversion in 3 hours.

Technical Relevance This patent is highly relevant because it starts from a true NBR latex composition and demonstrates direct post-hydrogenation in water. While it produces HNBR rather than low-ACN NBR, it is important for understanding how low-ACN NBR latexes can be processed and modified without solvent.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2238176B1 and US9657113B2 both start from NBR latex, but EP2238176B1 focuses on coagulation/isolation and impurity control, whereas US9657113B2 focuses on direct hydrogenation of the latex.
- EP2238176B1 discloses acrylonitrile/butadiene NBR with very low chlorine content (4–25 ppm), while US9657113B2 uses a latex containing 66% butadiene and 34% acrylonitrile; neither patent targets very low ACN explicitly, but both are composition-relevant to NBR processing.
- EP3063192B1 uses much higher nitrile contents (10–60 wt%, preferably 34–43 wt%) and is therefore closer to HNBR-type chemistry than low-ACN NBR; it adds 1–9 wt% acrylate termonomer and requires 50%+ hydrogenation.
- US7592409B2 is the only patent in the set that explicitly teaches controlled emulsion polymerization and in-situ emulsifier formation, making it the most process-relevant for composition-controlled low-ACN NBR synthesis.
- US9657113B2 demonstrates that NBR latexes can be hydrogenated efficiently at 100–120°C and 500–1500 psi H₂ with Ru/Os catalysts, but this is a downstream modification route rather than a synthesis route for low-ACN NBR.
- EP2238176B1 emphasizes PVA-assisted coagulation with MgCl₂ at 20°C followed by heating to 90°C, showing a low-salt isolation strategy that may help preserve rubber quality after emulsion polymerization.
- US7592409B2 uses 65–75°C microemulsion polymerization with potassium persulfate, tripotassium phosphate, and potassium hydroxide, illustrating a controlled aqueous polymerization platform that could be adapted to acrylonitrile/butadiene feed control.

4. CONCLUSION

The evidence indicates that low-acrylonitrile NBR is most plausibly produced by emulsion polymerization with tight monomer-feed/composition control, followed by low-contamination coagulation and drying. Among the patents reviewed, US7592409B2 provides the strongest general synthesis platform for controlled aqueous polymerization, while EP2238176B1 provides the most directly relevant NBR isolation/coagulation strategy and US9657113B2 confirms that NBR latexes with substantial acrylonitrile content can be processed in water without solvent. EP3063192B1 is more relevant as a nitrile-elastomer composition reference than as a low-ACN

NBR route. Based on the evidence, a practical low-ACN NBR process would favor emulsion polymerization of butadiene/acrylonitrile with controlled feed, moderate temperatures in the 65–75°C range, and coagulation using polyvalent salts with PVA or similar co-precipitants to minimize electrolyte demand and preserve vulcanization performance.

5. REFERENCES

- EP2238176B1 | Nitrile rubbers which optionally contain alkylthio terminal groups and which ... | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2238176B1/en>
- EP3063192B1 | Copolymer rubber containing nitrile groups | Not disclosed | EP | 2018 | <https://patents.google.com/patent/EP3063192B1/en>
- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
- US9657113B2 | Hydrogenation of a diene-based polymer latex | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9657113B2/en>